

Ankush Rai

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SUMMARY

Software Engineer specializing in scalable backend systems, distributed systems, and applied AI. Experienced shipping LLM/RAG/MCP-powered products and multi-tenant data integrations on AWS for Fortune 100 enterprise clients. Currently building event-driven ingestion infrastructure at Allyvia, taking systems from prototype to production.

EDUCATION

Santa Clara University

Master of Science in Computer Science and Engineering

Santa Clara, CA

Expected December 2026

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

India

June 2020

EXPERIENCE

Allyvia

Software Engineer Intern

March 2026 – Present

San Francisco (Remote)

- Built an event-driven client data ingestion pipeline using AWS SQS to decouple third-party sync jobs from the API, enabling reliable retries and back-pressure handling for high-volume QuickBooks/Square webhook traffic.
- Engineered a multi-tenant Python/Django REST API that integrates with QuickBooks and Square, unifying finance, inventory, and operations data into a single AI-driven ERP hub for SMB customers.

ZS Associates

Senior Software Engineer

March 2022 – January 2025

India

- Led migration from a Tableau dashboard to a custom React + FastAPI web app, shipping a platform that reduced load time by **40%** and eliminated **\$100K+** in licensing costs.
- Scaled FastAPI backend services with Redis caching and PostgreSQL pooling, instrumenting GitLab CI/CD, Pytest, and Prometheus for production monitoring, cutting P95 latency by **45%**.
- Designed and deployed an LLM-powered workflow on AWS for Pfizer (Fortune 100 pharma client), evaluating models and prompts that automated extraction from **10K+** reports, saving **20+** analyst-hours per week.

EY

Software Engineer I

October 2020 – February 2022

India

- Eliminated race conditions in concurrent Python services using threading locks, reducing transaction errors by **15%** on the General Motors OnStar production platform.
- Optimized response efficiency across Flask and Express.js applications using pagination and in-memory caching, improving API response time by **35%** for **10K+** monthly requests.

PROJECTS

PaperTrail: AI Document Intelligence | *Python, FastAPI, LangChain, Docker, AWS*

- Engineered an LLM-powered platform that processes unstructured PDFs into structured data with OCR, semantic chunking, and vector embeddings, automating manual document analysis. Won NVIDIA's "Agents for Impact" hackathon.

Argus: MCP Prompt-Injection Firewall | *Python, FastAPI, MCP, JWT, Claude-as-Judge, WebSockets*

- Designed a bidirectional MCP proxy with layered defense (regex + LLM-as-judge) and JWT-based per-agent scope policy that intercepts tool calls/responses, blocks prompt injection and credential exfiltration, and streams events to a live dashboard.

FormWhisper: Voice-Driven AI Tool | *Python, FastAPI, React 19, TypeScript, LangGraph*

- Shipped a full-stack AI tool that reads PDF forms using a Vision-Language Model (Qwen2.5-VL) and fills them through voice conversation. Won Best Future Unicorn at Hack for Humanity 2026 from **80+** teams.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frontend: React, Next.js, TypeScript, HTML/CSS, Responsive UI

Backend: Django, FastAPI, Flask, Express.js, REST APIs, Microservices

AI/ML: LLMs, RAG, AI Agents, LangChain, LangGraph, Prompt Engineering, scikit-learn, PyTorch

Databases: PostgreSQL, MySQL, Redis, Vector Databases

Cloud & DevOps: AWS (SQS, EC2, S3), Docker, Kubernetes, Kafka, GitLab CI/CD, Git, Prometheus, Pytest

AI Coding Tools: Claude Code (advanced), Cursor, GitHub Copilot

AWARDS

Spot Award (ZS Associates) – Recognized for shipping a Python LLM-powered workflow for Pfizer that automated manual pharma report analysis.

Best Future Unicorn – Hack for Humanity 2026, selected from **80+** teams for FormWhisper, an AI voice-driven form-filling tool.